A Mutli-Modal Approach To Human Action Anticipation

***Abstract*—This paper proposes a simple modification that can be made to existing human skeleton action recognition models that can improve action classification accuracy and early action recognition capability. We propose a method of extracting human early visual motion cues by sampling windows of optical flow surrounding skeleton keypoints in image space. Optical flow windows embed motion information in the learning process and enable the recognition of actions with fewer observed frames. With the addition of a lightweight embedding layer to learn windows of optical flow, three state of the art skeleton-based action recognition models (InfoGCN++, MS-G3D, ST-GCN++) show significant improvements in action classification accuracy and predictive ability. Evaluation on a fourth, RGB only dataset demonstrates the versatility of the InflowGCN++ model. Tests of embedding methods show greatest improvements in action classification accuracy using a CNN based architecture for learning optical flow windows.**

***Index Terms*—Human Action Recognition, Online Action Recognition, Human Motion Prediction, Multi-modal Media Computing**

I. Introduction

Human early action recognition has applications in fields such as human-robot teaming [15], pedestrian avoidance [10], abnormal activity detection in surveillance [11] and fall detection in aged care settings [3]. The task of early action recognition focuses on both the accuracy and speed of action classification [5]. This is particularly useful in real-time applications, such as robotics, where robot response is time-critical.

Human skeleton keypoints provide meaningful and succinct information about human actions. Skeleton-based action recognition has grown as an area of interest in the last two decades. The growing interest is partially due to the increase in availability of RGB-D sensors, giving rise to more skeleton-based action recognition datasets. It is also partially due to the simplicity of the input data; human skeleton keypoints contain *only* information about the position and movement of humans and lack noisy background elements.

However, skeleton-only approaches only capture relations between limbs, and do not cope well with limb occlusion. Skeleton action recognition datasets, such as NTU RGB+D [12], NTU RGB+D 120 [6] and NW-UCLA [14] use multiple RGB-D sensors to capture accurate human pose data from different viewpoints. These skeleton datasets fail to capture fine joint features that may distinguish similar actions.

RGB-only approaches to human action recognition must contend with large amounts of calculations needed to process video data. Complex environments, such as outdoor settings or shifting camera perspectives, present challenges for RGB-only models in capturing robust human action features [14]. RGB-only models designed to perform human early action recognition suffer when distinguishing intra-class differences and inter-class similarities.

To address these challenged, we propose a method of RGB pixel motion information into existing skeleton-based action recognition models. Our method, we have dubbed "FlowPose", involves the strategic sampling of windows of optical flow surrounding human skeleton keypoints. Optical flow can provide more nuanced information that the simple displacement vectors of skeleton keypoints. A simple embedding layer added to state-of-the-art skeleton-based action recognition algorithms shows an improvement in classification accuracy and action anticipation.

The main contributions of the proposed method are as follows:

- A feature extraction method for strategically sampling the optical flow field surrounding human pose keypoints.
- A simple embedding layer that can be easily implemented in existing skeleton-based action recognition models to learn windows of optical flow.
- An evaluation of three state-of-the-art skeleton-based action recognition models modified to learn additional optical flow features.
- Experimental results demonstrate outstanding performance on both skeleton-based and RGB-based human action recognition datasets for the purpose of human early action recognition.

II. Related work

Many of the state-of-the-art methods of learning human actions in sequences of skeleton keypoints is through graph convolution. In machine learning algorithm contexts, human skeleton keypoints depicted as graphs, with graph nodes and edges representing human joints and bones respectively. This compact representation lends itself well to machine learning techniques based in graph theory, such as graph convolutional networks (GCNs).

A GCN proposed by Duan et al. [4], ST-GCN++, achieved strong performance on skeleton-based action recognition datasets without the need for complicated attention mechanisms to model temporal dynamics. ST-GCN++ was designed to provide a strong baseline for future skeleton-based action recognition models. Liu et al. [7] designed MS-G3D,

Funding provided by the Australian Government and Swinburne University of Technology.

a skeleton-based action recognition model utilising spatial-temporal graph convolution modules to pass information across different graph scales.

Existing human-action recognition models must capture the entire action sequence in order to make predictions, which may present latency issues for real-time applications such as human-robot interaction. Chi et al. [2] incorporated a Neural Ordinary Differential Equations module into a skeleton-based action recognition model to model future skeleton states. The complete model, InfoGCN++, was able to classify human actions in skeleton datasets with fewer observed frames than existing skeleton-based action recognition models.

Multi-modal approaches to human action recognition area common. Cai et al. [1] proposed a two-stream architecture for hockey action recognition that utilised both optical flow and human poses. The inclusion of an optical flow stream increased hockey action classification by 10%.

III. Methods

We make a simple modification to three state-of-the-art skeleton-keypoint action recognition models to incorporate learnable early visual motion cues. The embedding layer of each model was modified to include two convolutional layers for learning optical flow windows. These learned features were then concatenated with the original skeleton keypoints coordinates. The models tested using this methodology were InfoGCN++ [2], ST-GCN++ [4] and MS-G3D [7].

The models were trained using hyperparametes defined by the authors. To test the model's performance on early action recognition, we utilised the last frame duplication strategy defined in [9]. Each model was trained on full observations of input sequences and tested using masking of test sequences at 20% observation ratios.

A. flow-pose

We have called our method of extracting early visual motion cues from videos "FlowPose". This is a general term for the method of sampling windows of optical flow surrounding human-pose keypoints. Based on our testing, a window size of 5×5 flow vectors demonstrated a good balance data efficiency and learnable features.

Optical flow provides richer information than joint displacement vectors. Where displacement indicate where a joint has come from, windows of optical flow can contain a more nuanced motion information about the moving pixels

Optical flow was extracted using the Pytorch implementation of the RAFT algorithm of optical flow estimation [13]. The UCF-101 dataset is RGB only. We extract human pose keypoints using the YOLO-POSE model [8].

IV. Results

The inclusion of learnable optical flow windows yields higher classification accuracy with the three models tested. Initial results indicate that our model performs better in most tasks. Observation accuracy is reported as area under the curve (AUC). For a given input sequence of length T, the model produces a class prediction for each frame T.

Attention graphs show which limbs our model is attending to.

V. Discussion

2 demonstrates that our model produces more accurate results with fewer frames. What this figure tells us is that our model, that utilises early visual motion cues, would only need to observe roughly 50% of the input sequence to produce a classification accuracy comparable to that of the base model. This is more dramatic in the NTU RGB+D 120 dataset, which might indicate that our model is able to learn more complex features from a more complex dataset. 3

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Fig. 1. Visual depiction of optical flow vectors sampled surrounding the skeleton keypoints. Here, we have averaged the window of optical flow vectors.

		NTU		NTU120		UCF-101		
		CS	CV	CSub	CSec	1	2	3
InfoGCN++	Base							
	CNN							
MS-G3D	Base							
	CNN							
ST-GCN++	Base							
	CNN							

Fig. 2. Classification accuracy (AUC) across observation ratios.

Fig. 3.

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Appendix

We tested three different methods of embedding the optical flow:

Absolute value	Taking the absolute value of the window of optical flow, producing a simple magnitude representing the *amount* of estimated motion surrounding the pose keypoint	IMG
Average value	Taking the average x and y values of the estimated optical flow, producing a vector representing the *direction* of estimated motion surrounding the pose keypoint	IMG
CNN learning	Learning the window of optical flow with two layers of convolutions	IMG